ELEC4631 Lecture 7: Stability, controllability and observability for LTI systems

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Laplace transform of matrix exponential function

- Let A be a n x n matrix and $f(t) = e^{At}$. We can write $f(t) = [f_{ij}(t)]_{i,j=1,2,..,n}$, where each $f_{ij}(t)$ is a scalar function of time.
- Laplace transform $F(s)$ of $f(t)$ determined by taking Laplace transform of each component of $f(t)$.
- That is, $F(s) = [F_{ij}(s)]_{i,j=1,2,..,n}$ where $F_{ij}(s)$ is Laplace transform of $f_{ij}(t)$.

Laplace transform of matrix exponential function

- Laplace transform of $f(t) = e^{At}$ can be computed as

$$
\begin{aligned}
F(s) &= \int_0^\infty e^{-st} e^{At}\, dt \\
&= \int_0^\infty e^{-sI_{n\times n}t} e^{At}\, dt \\
&= \int_0^\infty e^{(A-sI_{n\times n})t}\, dt \\
&= (A - sI_{n\times n})^{-1} \int_0^\infty (A - sI_{n\times n}) e^{(A-sI_{n\times n})t}\, dt \\
&= (A - sI_{n\times n})^{-1} \left[e^{(A-sI_{n\times n})t} \right]_0^\infty \\
&= (sI_{n\times n} - A)^{-1}
\end{aligned}
$$

Only valid for values of s for which this integral exists

Extended to all s in the complex plane that is not an eigenvalue of A via analytic continuation

Determining e^{Jt} for Jordan block J

- Let J be a single Jordan block with c on the diagonal.
- Laplace transform $F(s)$ of $f(t) = e^{Jt}$ is

$$F(s) = (sI - J)^{-1}$$

$$= \begin{bmatrix} s-c & -1 & 0 & 0 & \dots & 0 \\ 0 & s-c & -1 & 0 & \dots & 0 \\ 0 & 0 & s-c & -1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & s-c & -1 \\ 0 & 0 & 0 & \dots & 0 & s-c \end{bmatrix}^{-1} = \begin{bmatrix} (s-c)^{-1} & (s-c)^{-2} & (s-c)^{-3} & (s-c)^{-4} & \dots & (s-c)^{-n} \\ 0 & (s-c)^{-1} & (s-c)^{-2} & (s-c)^{-3} & \dots & (s-c)^{-n+1} \\ 0 & 0 & (s-c)^{-1} & (s-c)^{-2} & \dots & (s-c)^{-n+2} \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & (s-c)^{-1} & (s-c)^{-2} \\ 0 & 0 & 0 & \dots & 0 & (s-c)^{-1} \end{bmatrix}$$

Determining e^{Jt} for Jordan block J

- $F(s)$ can thus be expressed as

$$F(s) = (s-c)^{-1} I + (s-c)^{-2} F_1 + (s-c)^{-3} F_2 + \ldots + (s-c)^{-n} F_{n-1}$$

with F_j a matrix with ones in the j-th upper diagonal (the 0-th upper diagonal being the diagonal) and zeros everywhere else.

Determining e^{Jt} for Jordan block J

- Taking inverse Laplace transform of each term on right hand side gives

$$
\begin{aligned}
e^{Jt} &= e^{ct}I + te^{ct}F_1 + \frac{t^2}{2!}e^{ct}F_2 + \ldots + \frac{t^{n-1}}{(n-1)!}e^{ct}F_{n-1} \\
&= e^{ct}\left(I + tF_1 + \frac{t^2}{2!}F_2 + \ldots + \frac{t^{n-1}}{(n-1)!}F_{n-1}\right) \\
&= e^{ct}\begin{bmatrix}
1 & t & \frac{t^2}{2!} & \frac{t^3}{3!} & \ldots & \frac{t^{n-1}}{(n-1)!} \\
0 & 1 & t & \frac{t^2}{2!} & \ldots & \frac{t^{n-2}}{(n-2)!} \\
0 & 0 & 1 & t & \ldots & \frac{t^{n-3}}{(n-3)!} \\
\vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\
0 & 0 & \ldots & 0 & 1 & t \\
0 & 0 & \ldots & 0 & 0 & 1
\end{bmatrix}
\end{aligned}
$$

Internal stability of LTI state-space systems

- An LTI state-space system is (*internally*) *stable* if for each x_0 there is a (finite) constant $C(x_0) > 0$, dependent on x_0, such that
$$\|x(t)\| \leq C(x_0) \quad \forall t \quad \geq 0$$
when there is no input ($u(t) = 0$ for all $t \geq 0$) .
- An LTI state-space system is (*internally*) *asymptotically stable* if for any initial condition x_0
$$\lim_{t\to\infty} x(t) = 0$$
when there is no input ($u(t) = 0$ for all $t \geq 0$) .

Internal stability of LTI state-space systems

- An LTI state-space system is (*internally*) *marginally stable* if it is stable but <u>not</u> asymptotically stable.
- An LTI state-space system is (*internally*) *unstable* if $\|x(t)\| \to \infty$ as $t \to \infty$ for some initial state x_0.
- A stable system falls into one of two categories: either asymptotically stable or marginally stable.

Internal stability of LTI state-space systems

- Connections with concepts from Lecture 5 are:
  - Marginal stability of LTI state-space system implies that every equilibrium point of system is stable i.s.L.
  - Asymptotic stability of LTI state-space system implies that the origin is a globally asymptotically stable equilibrium point of system.
- We shall shortly see that unlike general nonlinear dynamical systems, stability of LTI state-space system is easy to verify.

LHP, RHP, and the imaginary axis

- The *open left half plane*, or simply LHP, is the set of all complex numbers with real part less than 0.
- The *open right half plane,* or simply RHP, is the set of all complex numbers with real part greater than 0.
- The boundary between LHP and RHP on complex plane is the imaginary axis.
- Note that neither LHP nor RHP include the imaginary axis.

Internal stability of LTI state-space systems

- Theorem 1: LTI state-space system is asymptotically stable <u>if and only if</u> all eigenvalues of A are in LHP.

- Theorem 2: LTI state-space system is marginally stable <u>if and only if</u> (i) A has eigenvalues on imaginary axis and all such eigenvalues have geometric multiplicity = algebraic multiplicity, and (ii) all other eigenvalues of A in LHP.

Internal stability of LTI state-space systems

- Theorem 3: LTI state-space system is unstable <u>if and only if</u> (i) there is an eigenvalue of *A* in RHP and/or (ii) on imaginary axis with geometric multiplicity < algebraic multiplicity.

Internal stability of LTI state-space systems

- Recall that any matrix A has Jordan block decomposition

Jordan blocks. Each J_k is a matrix placed along the diagonal block

$$A = V \begin{bmatrix} \lambda_1 & & & & & & \\ & \ddots & & & & & \\ & & \lambda_m & & & & \\ & & & J_{m+1} & & & \\ & & & & J_{m+2} & & \\ & & & & & \ddots & \\ & & & & & & J_{m+s} \end{bmatrix} V^{-1}$$

$$J_k = \begin{bmatrix} \lambda_k & 1 & & & \\ & \lambda_k & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_k & 1 \\ & & & & \lambda_k \end{bmatrix}$$

Internal stability of LTI state-space systems

- Whence

$$e^{At} = V \begin{bmatrix} e^{\lambda_1 t} & & & & & & \\ & \ddots & & & & & \\ & & e^{\lambda_m t} & & & & \\ & & & e^{J_{m+1} t} & & & \\ & & & & e^{J_{m+2} t} & & \\ & & & & & \ddots & \\ & & & & & & e^{J_{m+s} t} \end{bmatrix} V^{-1}$$

Internal stability of LTI state-space systems

- The theorems are true because

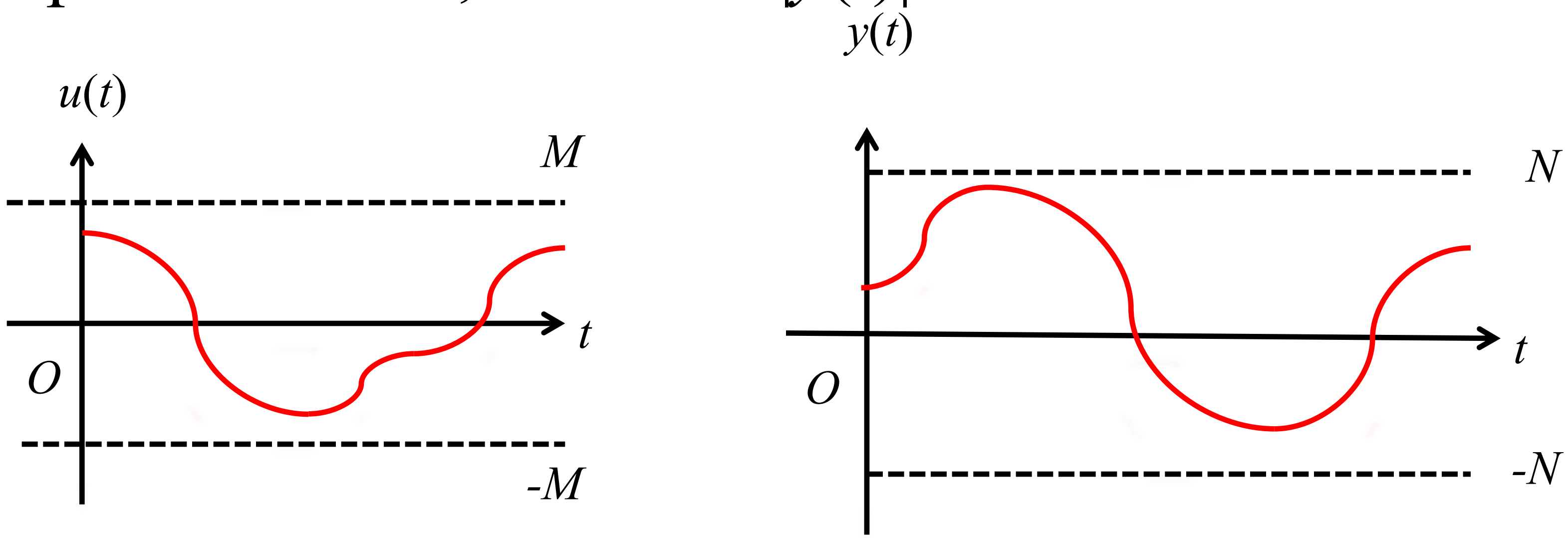
When all eigenvalues are in LHP (have negative real parts) all these terms go to 0 as $t \rightarrow \infty$

$$e^{At} = V \begin{bmatrix} e^{\lambda_1 t} & & & & & & \\ & \ddots & & & & & \\ & & e^{\lambda_m t} & & & & \\ & & & e^{J_{m+1} t} & & & \\ & & & & e^{J_{m+2} t} & & \\ & & & & & \ddots & \\ & & & & & & e^{J_{m+s} t} \end{bmatrix} V^{-1}$$

If there are eigenvalues in RHP or eigenvalues on imaginary axis with geometric multiplicity < algebraic multiplicity then some terms will blow up as $t \rightarrow \infty$.

BIBO stability

- A SISO LTI system is bounded input-bounded output (BIBO) stable if, *starting from initially at rest* ($x_0 =$ 0), given any input $u(t)$ with $|u(t)| \leq M$ for all $t \geq 0$ and some $M > 0$, there is another constant $N > 0$, dependent on M, such that $|y(t)| \leq N$ all $t \geq 0$.



BIBO stability from initially at rest

- Characterisation: A SISO LTI state-space system is BIBO stable if and only if all poles p of $H(s) = C(sI-A)^{-1}B + D$ such that $\lim_{s \to p} |H(s)| = \infty$ (i.e., pole p is not cancelled by a zero of $H(s)$) are in LHP.

BIBO stability

- If $x_0 = 0$ and

$$H(s) = \frac{s+4}{s^2+5s+6} = \frac{s+4}{(s+3)(s+2)}$$ (BIBO stable)

$$H(s) = \frac{s^2+s-2}{s^2+3s-4} = \frac{(s-1)(s+2)}{(s-1)(s+4)}$$ (BIBO stable)

$$H(s) = \frac{s^2+3s+2}{s^2+s-2} = \frac{(s+1)(s+2)}{(s-1)(s+2)}$$ (Not BIBO stable)

Asymptotic stability vs BIBO stability

- Notion of asymptotic stability stronger than BIBO stability→former guarantees latter but <u>not</u> vice-versa.
- BIBO stable system need not be asymptotically stable.
- Example: LTI state-space system below is BIBO stable but not asymptotically stable.

$$S = \left(\begin{bmatrix} -1 & 0 \\ -3 & 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 0 \right)$$

Asymptotic stability vs BIBO stability

$$S = \left(\begin{bmatrix} -1 & 0 \\ -3 & 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 0 \right)$$

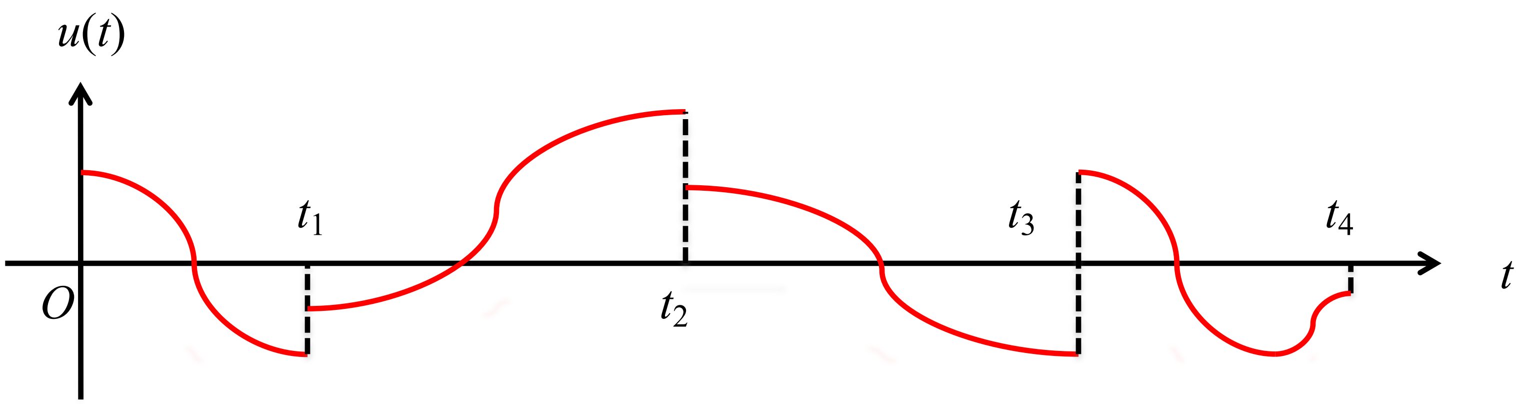
- Eigenvalues of A are -1 and 2, unstable.
- Solving state-space equation gives

$$y(t) = e^{-t} x_{10} + \int_0^t e^{-(t-s)} u(s)\, ds$$

so system is BIBO stable.

Piecewise continuous signals

- A signal $u(t)$ is *piecewise continuous* if there exists a sequence of increasing times $t_0 = 0 < t_1 < t_2 < \ldots < t_\mathrm{k} < \ldots$ with $\lim_{r\to\infty} t_r = \infty$ such that for each $k = 0, 1, 2, \ldots$ $u(t)$ is continuous in the semi-open interval $t_k \leq t < t_{k+1}$.



Controllability

- Suppose LTI state-space system starts at initial state x_0 and we want to move it to desired state x_f at some final time $t_f > 0$ (i.e., $\mathrm{x}(t_f) = x_f$).
- Is there a piecewise continuous control $u(t)$ that will drive state from x_0 at $t = 0$ to x_f at $t = t_f$?
- System is said to be *controllable* if for any initial state x_0, any final time $t_f > 0$, and any final state x_f, there is a piecewise continuous control $u(t)$, that depends on x_0, t_f, and x_f, that brings system state from x_0 to x_f at time t_f.

Observability

- Suppose LTI state-space system without input ($u(t) = 0$ for all $t \geq 0$) starts at an *unknown* initial state x_0, and output $y(t)$ can be observed from $t = 0$ to a final time $0 < t_f < \infty$.
- Given observations of $y(t)$ for $0 \leq t \leq t_f$ is it possible to determine x_0?
- System is said to be *observable* if given any unknown initial state x_0 and any final time $t_f > 0$, x_0 can be unambiguously determined, when there is no input, based on observations of $y(t)$ in the interval $0 \leq t \leq t_f$.

Controllability and observability matrix

- Introduce two special matrices, the *controllability matrix* $\mathscr{C}$ and *observability matrix* $\mathscr{O}$:

$$\mathscr{C} = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} \quad \mathscr{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Kalman's characterisation of controllability and observability

- Rudolf Kalman in the 60s proved the following remarkable fundamental results (but for general MIMO systems).
- Theorem (controllability): A SISO LTI state-space system is controllable if and only if the controllability matrix $\mathscr{C}$ is invertible.
- Theorem (observability): A SISO LTI state-space system is observable if and only if the observability matrix $\mathscr{O}$ is invertible.

Kalman's characterisation of controllability and observability

- Results are remarkable in that checking complicated notions of controllability and observability for SISO LTI state-space systems can be reduced to simply checking invertibility of certain matrices.

Example – simultaneously uncontrollable and unobservable system

- The following system in modal canonical form is *not* controllable and *not* observable.

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)$$

- Observability and controllability matrices are *not* invertible:

$$\mathscr{C} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}; \quad \mathscr{O} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

- Why? This example can be understood intuitively.

Some important remarks

- Controllability and observability are properties of a particular state-space representation/realisation, not of the system transfer function.
- State-space realisation of a given transfer function is not unique. A particular realisation could be controllable, observable, both, or even none. These properties are realisation dependent, not transfer function dependent.

Uncontrollable and unobservable systems

- The system

$$S = \left(\begin{bmatrix} 2 & 12 \\ -1 & -5 \end{bmatrix} \; , \; \begin{bmatrix} -3 \\ 1 \end{bmatrix} \; , \; \begin{bmatrix} 1 & 2 \end{bmatrix} \; , \; 0 \right)$$

  is not controllable but observable.

- Eigenvalues of A: -1 and -2.
- Transfer function:

$$H(s) = -\frac{\cancel{s+1}}{\cancel{(s+1)}(s+2)} = -\frac{1}{s+2}$$

Uncontrollable and unobservable systems

- The system

$$S = \left(\begin{bmatrix} 2 & -1 \\ 12 & -5 \end{bmatrix} \; , \; \begin{bmatrix} 1 \\ 2 \end{bmatrix} \; , \; \begin{bmatrix} -3 & 1 \end{bmatrix} \; , \; 0 \right)$$

  is not observable but controllable.
- Eigenvalues of A: -1 and -2.
- Transfer function:

$$H(s) = -\frac{\cancel{s+1}}{\cancel{(s+1)}(s+2)} = -\frac{1}{(s+2)}$$

Uncontrollable and unobservable systems

- Controllable and observable → no pole-zero cancellation can occur in transfer function.
- System uncontrollable and/or unobservable → pole-zero cancellation will occur in transfer function.

Uncontrollable and unobservable systems

- The second order differential system

$$\ddot{y}_1(t) - \dot{y}_1(t) - 2y_1(t) = \dot{u}(t) - 2u(t)$$

and first order system

$$\dot{y}_2(t) + y_2(t) = u(t)$$

have same transfer function: $Y_1(s)/U(s) = Y_2(s)/U(s)$ $1/(s+1)$. This is due to cancellation in the first system.

- But the nature of their respective solutions are very different!

Uncontrollable and unobservable systems

- Observer canonical state space representations:

$$\ddot{y}_1(t) - \dot{y}_1(t) - 2y_1(t) = \dot{u}(t) - 2u(t)$$

$$\Rightarrow S_1 = \left(\begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} , \begin{bmatrix} -2 \\ 1 \end{bmatrix} , \begin{bmatrix} 0 & 1 \end{bmatrix} , 0 \right)$$

and

$$\dot{y}_2(t) + y_2(t) = u(t) \Rightarrow S_2 = \left(-1 , 1 , 1 , 0 \right)$$

Uncontrollable and unobservable systems

- Solving state-space equations gives:

$$x_1(t) = \frac{1}{3}\begin{bmatrix} e^{2t} + 2e^{-t} & 2e^{2t} - 2e^{-t} \\ e^{2t} - e^{-t} & 2e^{2t} + e^{-t} \end{bmatrix} x_{10} + \int_0^t \begin{bmatrix} -2e^{-(t-s)} \\ e^{-(t-s)} \end{bmatrix} u(s)\,ds$$

$$y_1(t) = \frac{1}{3}\begin{bmatrix} e^{2t} - e^{-t} & 2e^{2t} + e^{-t} \end{bmatrix} x_{10} + \int_0^t e^{-(t-s)} u(s)\,ds$$

$$x_2(t) = x_{20} e^{-t} + \int_0^t e^{-(t-s)} u(s)\,ds$$

$$y_2(t) = x_{20} e^{-t} + \int_0^t e^{-(t-s)} u(s)\,ds$$

Uncontrollable and unobservable systems

- In general, $y_1(t) \neq y_2(t)$ for all $t > 0$ even if $y_1(0) = y_2(0)$.
- S_1 observable but not controllable. It is <u>not</u> internally stable as one eigenvalue in RHP but is BIBO stable.
- S_2 observable and controllable. S_2 is internally asymptotically stable and thus also BIBO stable.